

Research Article

Structural and Functional Characterization at the Molecular Level of the MATE Gene Family in Wheat in Silico

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A series of multidrug extrantransporters known as the multidrug and potentially toxic extrusion (MATE) genes are found in all living things and are crucial for the removal of heavy metal ions, metalloids, exogenous xenobiotics, endogenous secondary metabolites, and other toxic substances from the cells. However, there has only been a small amount of them in silico analysis of the MATE family of genes in plant species. In the current study, the MATE gene family was characterized in silico where two families and seven subfamilies based on their evolutionary relationships were proposed. Plant breeders may use TraesCS1D02G030400, TraesCS4B02G244400, and TraesCS1A02G029900 genes for marker-assisted or transgenic breeding to develop novel cultivars since these genes have been hypothesized from protein-protein interaction study to play a critical role in the transport of toxic chemicals across cells. The exon number varies from 01 to 14. One exon has TraesCS1A02G188100, TraesCS5B02G562500, TraesCS6A02G256400, and TraesCS6D02G384300 genes, while 14 exons have only two genes that are TraesCS6A02G418800 and TraesCS6D02G407900. Biological stress (infestations of disease) affects the expression of most of the MATE genes, with the gene TraesCS5D02G355500 having the highest expression level in the wheat expression browser tool. Using the Grain interpretation search engine tool, it is found that the vast bulk of MATE genes are voiced throughout biotic environmental stresses caused by disease pests, with the genotype TraesCS5B02G326600.1 from family 1 exhibiting the greatest level of expression throughout *Fusarium* head blight infection by *Fusarium graminearum* after 4 days of infection. The researchers constructed 39 ternary plots, each with a distinct degree of expression under biotic and abiotic stress settings, and observed that 44% of the triplets have imbalanced outputs (extreme values) due to their higher tissue specificity and increased intensity.

1. Introduction

Bread wheat (*Triticum aestivum* L.), a significant cereal crop, provides around 35% of the world's major diet [1]. This plant has been identified as the youngest polyploid species [2].

Each of the three genomes, A, B, and D, contains seven chromosomes [1]. *Triticum urartu*, *Aegilops speltoides*, and *Aegilops tauschii* are the three diploid progenitors of these three genomes, and each of these genomes has a unique set of characteristics, such as gene density, functional and

expression variation, and epistasis [3, 4]. Five times higher than humans, eight times higher than maize, and forty times higher than rice, the hexaploid wheat genome is around 17 Gb in size [5, 6]. To deal with a wide range of toxic compounds, plants have evolved a sophisticated detoxification system, including weighty metals, metalloids, exogenic xenobiotics, and endogenic secondary phospholipids such as atropine and saponins, terpenoids, terpenoid-obtained auxins, skin lipids, and monolignols, as a result of environmental pressures [7, 8]. Following previous studies, these compounds are expelled from plant cells via postenzymatic modifications or accumulate in the apoplastic dungeon walls or middle germs. In the genome of plants, MATEs often comprise a large family. For functional differentiation, homologous MATE transporters have various subcellular localized versions, substrate preferences, and reactions to environmental stimuli. Plants' polyphenol, alkaloids, plant hormones, and iron chelating agents are examples of the substrates for MATEs. The buildup of these compounds is frequently linked to beneficial agronomic properties including the colors of the seeds and fruits, the harmony between seed dormancy, flavor, and resilience to stress. Wild and domesticated germplasms typically differ in agronomic characteristics including seed color, seed flavor, and stress tolerance in plants. Incredibly, the MATE genes are associated with an antiporter/xenobiotic transmembrane transporter family of proteins. One of the most well-known transporter families in plants, this gene family has many members [9, 10]. Organic acids, for example, are transported by this family of transporters during development and play an essential role in many physiological processes. The physiological functions, tissue specificity, membrane location, and affinity for diverse substrates can all be used to classify MATE transporters [10]. MATE transporters carry a wide range of substrates, including organic acids, plant hormones, and secondary metabolites, and are engaged in a wide range of physiological processes during plant development. The multigene family of multidrug and potentially toxic extrusion (MATE) transporters mediates several tasks in plants by ejecting a variety of substrates, such as chemical molecules, specialized metabolites, hormones, and xenobiotics. MATE categorization based on genome-wide investigations is still unclear, most likely because there are not enough extensive phylogenomic studies and/or databases of reference sequences. The MATE transporters were discovered as multidrug efflux proteins in *Vibrio parahaemolyticus* and *Escherichia coli* and were named after their biological function because of the shortage of ordered similarity with other known transporters [11, 12]. *Arabidopsis thaliana*'s AtALF5 plant-type MATE transporter was discovered to be actively involved in multidrug resistance [13]. As demonstrated in several plant species, including rice and *Arabidopsis* [14], tomato [15], soybean [16], cotton [17], *Sorghum bicolor* [8], blueberry [18], and *Camellia sinensis* [19], the MATE gene family may aid in the fight against sustainability threats associated with the current agro-climatic situation, such as the presence of heavy metals in soil and the alteration of the chemical properties of topsoil. To identify the MATE gene family and apply them in the creation of gene-based

molecular creator of herb breeding, the wheat whole-genome pattern in 2018 demands gene annotation, particularly in silico. The current study obtained wheat MATE gene sequences and looked at how they linked with other proteins. Additionally, we investigated the in silico expression of MATE genes with corresponding homoeologous candidates under various biotic and abiotic stress situations. Our study will improve the scientific understanding of the structure, function, and expression of plant MATE genes.

2. Materials and Methods

2.1. Phylogenetic Classification and Multiple Sequence Alignment Analysis of Wheat MATE Proteins. The use of keyword (query)-based search over homology-based sequence similarity was adopted here since false negatives (homologs with nonsignificant scores) cannot be determined when applying tools such as BLAST, FASTA, and HMMER for the discovery of the gene. Moreover, DNA:DNA alignment statistics are less precise since their expected values of 10–6 often occur by accident, while 10–10 is a more widely recognized limit for homology based on DNA:DNA searches. So, it was suggested from a previous study to copy the name and function of the reference protein to the query sequence when looking for genes in a database. This is because the information on modified residues, active sites, variation, and mutation studies make it easier to figure out how evolution works to strengthen the power of query-based search [20–22]. The homology-based search could have found a lot of protein-coding sequences. But text-based search only found valid and documented MATE genes, which is what we were looking for. These data were obtained from the following sites accessible in the public domain: <https://www.uniprot.org/>, <https://www.uniprot.org/help/uniparc>, and <http://www.gramene.org/>. The UniProt database of protein resources is a comprehensive, high-quality, and open resource for all proteins worldwide. To reliably identify the same protein across several databases, UniPrac uses a single, nonredundant database to contain unique sequences and provide a stable and unique identifier [23]. The “Gramene” data repository was used to create this karyotypic chromosomal overview [24]. The phylogenetic analysis of MATE peptide sequences was performed in W-IQ-TREE (<http://iqtree.cibiv.univie.ac.at/>) using 1,000 ultrabootstraps as the default [25, 26]. A new tree was also created at <http://www.phylogeny.fr/onetask>. A new phylogenetic tree was also created at Phylogeny.fr (<http://www.phylogeny.fr/phylo.cgi/treedyn.cgi>) from the consensus Newick format calculated in W-IQ-TREE [27] to improve visualization. to improve visualization. Forty-four MATE protein sequences were aligned according to their phylogenetic grouping using the constraint-based multiple alignment tool [28, 29]. Multiple sequence alignment (MSA) is used to find common patterns and evolutionary links among genes. It specifically refers to the alignment of at least three biological sequences, most commonly DNA, RNA, or protein. Algorithms for computing produce and analyze alignments, whereas a phylogenetic alignment is different from other types of multiple sequence alignment because it

must align homologous characteristics. In order for the aligned sequences to correctly represent the events represented by the homologies, the alignment procedure's objective should be to identify the events connected to the homologies.

2.2. Protein-Protein Interaction among MATE Proteins and Gene Co-occurrence Study. Using the STRING (search tool for retrieval of interacting genes/proteins) online tool, we produced information about the functional association by combining known and projected protein-protein linkages for several species where indirect functional interactions were also considered [30, 31]. According to network data, 27 nodes in the network of proteins were found in functional subsystems, with an average node degree of 0.148 and PPI enrichment p -value of 1.9×10^{-8} . The “high confidence” cut-off of 0.700 was used while screening interaction linkages [32]. If the “MORE” option had been used in the STRING interface, we could have obtained more interactions, but this was not our primary goal in this study, which focused on the protein-protein interactions of the selected MATE proteins.

2.3. In Silico Expressions and Ternary Plots Analysis. Predicting gene expression levels from wheat RNA-sequence data sets was done using the expression visualization and integration online platform expVIP (<http://www.wheat-expression.com>). To better understand how MATE genes might be expressed under different situations, we used a polyploid wheat gene expression database. At various phases of development in wheat, the method is employed to know homologous specific transcript profiling [33, 34]. Several factors were considered while setting parameters for the current experiment, including previous study, tissues, age, stress, and variety. The results were based on high tissue expression levels, such as cereals, roots, leaves or shoots, vertebrae, and pong. HEATMAP was used to illustrate the wheat MATE gene expression potential in response to various biotic and abiotic stress situations. $\log_2$ (TPM) expression values were used with normalization to construct the HEATMAP. EnsemblPlant and the wheat expression browser were used to find wheat MATE gene homoeologs. The wheat expression browser also shows the potential homologous possibilities for 44 MATE genes in ternary plots. According to their positions in the ternary plot for examining individual tissues and different stress circumstances, each homoeolog's comparative expressions were calculated.

3. Results

3.1. Phylogenetic Classification and Multiple Sequence Alignment Analysis of Wheat MATE Proteins. Bread wheat contains an estimated 107,891 coding genes and 12,853 noncoding genes (Appels et al. 2018). A total of 44 MATE genetics had been discovered, and their relevant sequences were downloaded from the Gramene data repository. According to Figure 1, the MATE gene family was divided into two core groups or families on a scale of 1, and seven

minor groups or subfamilies were proposed on a scale of 2.57. Two subfamilies were found in the first family, whereas the second family included five subfamilies. Further in-depth analysis of the reasons for the genetic variation across genes with comparable functions and biological activities was carried out using multiple sequence alignment of those protein sequences [35, 36]. The phylogenetic grouping of 44 MATE protein sequences was produced from their protein sequence alignment. Wheat MATE genes' structural diversity and functional similarity were the primary focus of research on the number of exons and introns in each gene. The exon number varies from 01 to 14. The most numbers of the genes have 07 to 08 exons, one exon has 04 genes, among that TraesCS6A02G418800 and TraesCS6D02G407900 have the greatest number of exons of 14. TraesCS1A02G188100, TraesCS5B02G562500, TraesCS6A02G256400, and TraesCS6D02G384300, on the other hand, had the smallest number of exons of 01. The structural analysis provided further proof beyond phylogenetic studies for their sequence variation at the genomic level while maintaining the same functionality.

3.2. Protein-Protein Interaction among MATE Proteins and Gene Co-occurrence Study. It is critical to identify the protein-level functional relationship or joint biological function of wheat MATE genes [37]. The STRING software itself identified just three MATE proteins W5ANP1 (TraesCS1D02G030400), W5E135 (TraesCS4B02G244400), and W4ZYL7 (TraesCS1A02G029900) as having substantial interactions among themselves, out of the 27 involved proteins (Figure 2).

Enodulin 93 (ENOD93) like proteins were found to be functionally linked with these three proteins W5ANP1, W5E135, and W4ZYL7 (Figure 3).

The co-occurrence patterns of 27 involved MATE proteins were also examined to understand better how these proteins interact with each other across various taxa (Figure 4).

3.3. In Silico Expressions and Ternary Plot Analysis. Like other plants, the wheat genome contains MATE transporter genes that control various expressions and operations throughout vegetative growth, reproductive improvements, senescence, and obstruction to biotic and abiotic stressors. Leaf, root, shoot, flower, grain, spike, and other tissues were the most abundant sources of MATE gene transcripts. Genes associated with stress response were found to be expressed in a wide range of environments. However, most genes are activated under biotic stress, resulting from stresses. The study of biotic and abiotic stressors occurring at the same time reveals the intricate processes by which plants adapt their response to specific environmental circumstances. The observation of a novel transcriptional pattern that was distinct from either stress alone supports the idea that a possibly more harmful abiotic stress can overturn a potentially less harmful biotic stress. Under *Fusarium* head blight, TraesCS5B02G326600 has the maximum expression level. However, the sites of expressions were spikes, spikelets,

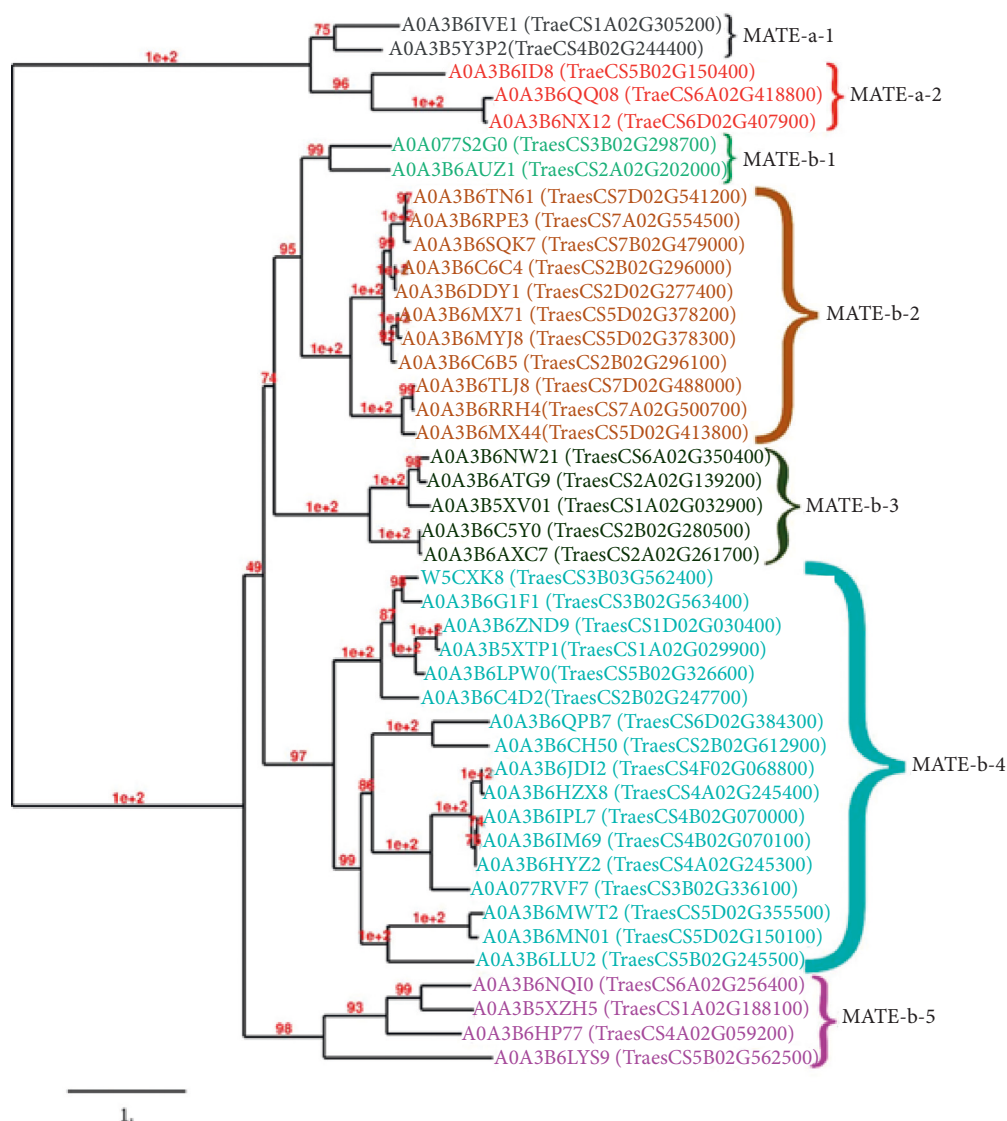


FIGURE 1: Phylogeny of 44 MATE genes classified into 2 major/primary groups or families at a scale of 1 and 7 secondary groups/subfamilies were proposed at a scale of 2.57.

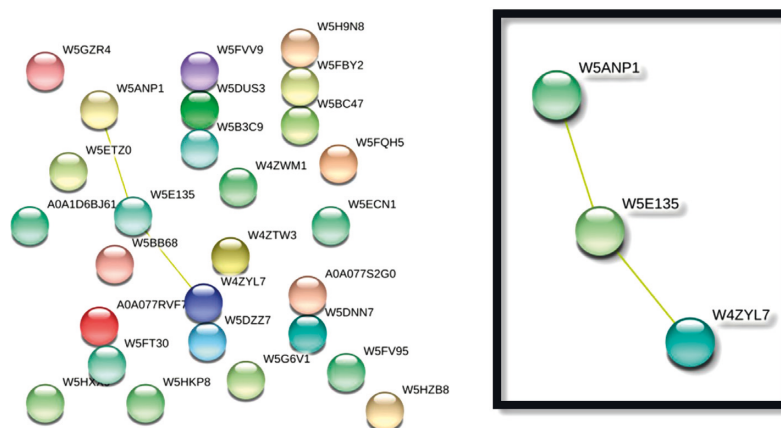
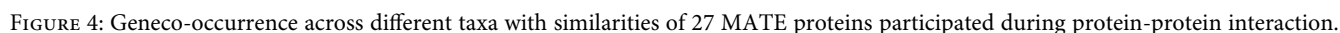
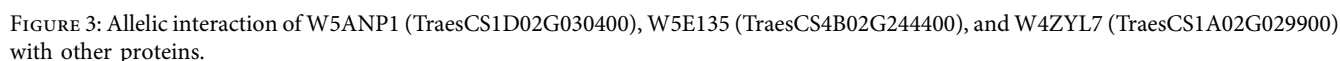


FIGURE 2: Protein-Protein interaction among 44 wheat MATE proteins.



there. Genes from the same family did not have similar patterns of expression, which was unexpected. TraesCS1A02G305200, TraesCS2B02G247700, TraesCS2D

02G277400, TraesCS3B02G298700, TraesCS4B02G244400, TraesCS5B02G326600, and TraesCS2B02G296000 all showed low to moderate expression when subjected to abiotic stresses. The highest level of TraesCS3A02G265100 gene expression was found with TraesCS3B02G298700 at 63.62%. In comparison to its counterparts, TraesCS5D02G355500 had the greatest level of expression (69.79%) under biotic stress. Also, we found that all the genes were expressed at full potential during stressed conditions. Genes such as TraesCS1B02G315900, TraesCS7D02G488000, TraesCS2A02G222300, TraesCS2D02G277400, TraesCS5B02G371200, TraesCS7D02G488000, TraesCS2D02G277400, TraesCS1A02G305200, and TraesCS7A02G500700 are examples of homologs (Figure 5). Thirty-nine ternary graphs containing simultaneous homologous genes were identified for the 44 MATE genes, revealing significantly various expression stages under biotic and abiotic conditions for 39 genes, with 56% of triads (A, B, and D homeologs) exhibiting equal expression and 44% triads exhibiting nonbalanced expression, indicating greater tissue specificity and expressivity under stress conditions. Each circle dot represents a gene trio, with A, B, and D coordinates indicating how much each homeolog contributes to the total expression of the triad [38, 39]. Given wheat's repeated susceptibility to many diseases, these newly discovered MATE genes may be crucial for producing more disease-tolerant cultivars and deserve further study.

3.4. Discovery of Protein Motifs in Different Subfamilies. Short amino acid configurations are known as “motifs” and are shared by members of the protein family. In order to attribute potential functions to unidentified proteins, they are intended to be utilized in combination with protein sequence databases. The goal of the motif is to identify patterns in the sequences of biopolymers (such as proteins or nucleotides) to comprehend the form and composition of the components the sequences represent. Phylogenetic subfamilies were used to classify MATE protein motifs. Earlier research revealed that the motif compositions of most closely related members of the same family were comparable, implying functional commonality [7, 40]. A total of 22 preserved motifs were detected on this occasion. The kinds and sequences of protein motifs differed significantly between the MATE-a-1, MATE-a-2, and MATE-b-1 subfamilies.

4. Discussion

In soybean and maize, 117 MATE genes [16] formed 4 core clades/families, while 49 ZmMATE genes [40] formed 7 clusters. A total of 196 putative MATE genes were investigated in the same way to reconstruct the evolutionary history of 2 cotton species and 1 *Arabidopsis* species, and the findings indicated 3 subfamilies based on phylogenetic classification: GaMATE, GrMATE, and *Arabidopsis* MATE. GaMATE had 68 genes; GrMATE comprised 70; and *Arabidopsis* MATE contained 58. Cotton MATE genes are highly conserved and

functionally diversified, as shown by phylogeny and intron-exon studies [5]. In today's study, functional similarity was observed in structural diversity. The classification of wheat genes was proposed without evaluating the in vivo or in vitro validity. The validity could not be assessed in this issue.

In situ hybridization method was employed in rice to examine the expression of the OsENOD93-1 gene in the roots of wild-type plants to understand in-depth cell-specific transcription. Therefore, ENOD93 was found to express in vascular bundles and the epidermal and endodermal layers, suggesting a possible role in transporting chemicals from the root to the shoot [41, 42]. In rice, MATE genes were also proposed to have a substantial role in removing a metalloid, arsenic from cells, ensuring the safety of the embryonic development inside the seed [43]. Henceforth, in the present investigation, genes encoding W5ANP1 (TraesCS1D02G030400), W5E135 (TraesCS4B02G244400), and W4ZYL7 (TraesCS1A02G029900) may play a critical role in transporting cellular toxic chemicals. Plant breeders may use genes TraesCS1D02G030400, TraesCS4B02G244400, and TraesCS1A02G029900 for marker-assisted or transgenic breeding to develop novel cultivars that can withstand unforeseen cellular toxic conditions.

MATE genes of *Hordeum Vulgare*, *Oryza sativa*, *Setaria italica*, *Zea mays*, *Musa acuminata*, and many other species had a high degree of genetic similarity with wheat MATE genes. Additionally, MATE-type proteins, which have a high level of genetic co-occurrence, suggest that the protein is an essential component of an organism's ability to survive [44, 45]. So MATE genes can be considered as a lifesaving genomic component that is structurally and functionally conserved across species [46]. The importance of these evolutionary signals in knowing the fundamental architect of wheat genes cannot be overstated. ClustalX was also utilized to evaluate the various pattern sequence of 08 ideal VcMATE protein in blueberry, as well as specific MATE transporter orthologs (*Vaccinium corymbosum*). VcMATE 2 showed the most striking resemblance to known flavonoid transporters, while VcMATE 1 and VcMATE 4 showed only a passing resemblance to MATE-type flavonoid transporters [18].

A series of epistatic interactions among genes have a critical role in the resulting phenotype of a trait. Gene mutations are seldom reflected at the phenotypic level, owing to the frequency and quantitative nature of allele interactions on the phenotype [47]. Allelic interactions occur at the protein level in the cellular environment. Several currently used computational techniques anticipate the functional impact of encoding human alleles using relative sequence-based analysis and study of protein structure. Investigation of human genetic variation can benefit from functional and structural analyses of coding allelic variations. It aids in evaluating the potency of genetic variation against harmful missense mutations and studies the influence of demographic history on harmful genetic variation in population and developmental genetics. It could help with the diagnosis of unidentified mutations in the genes causing

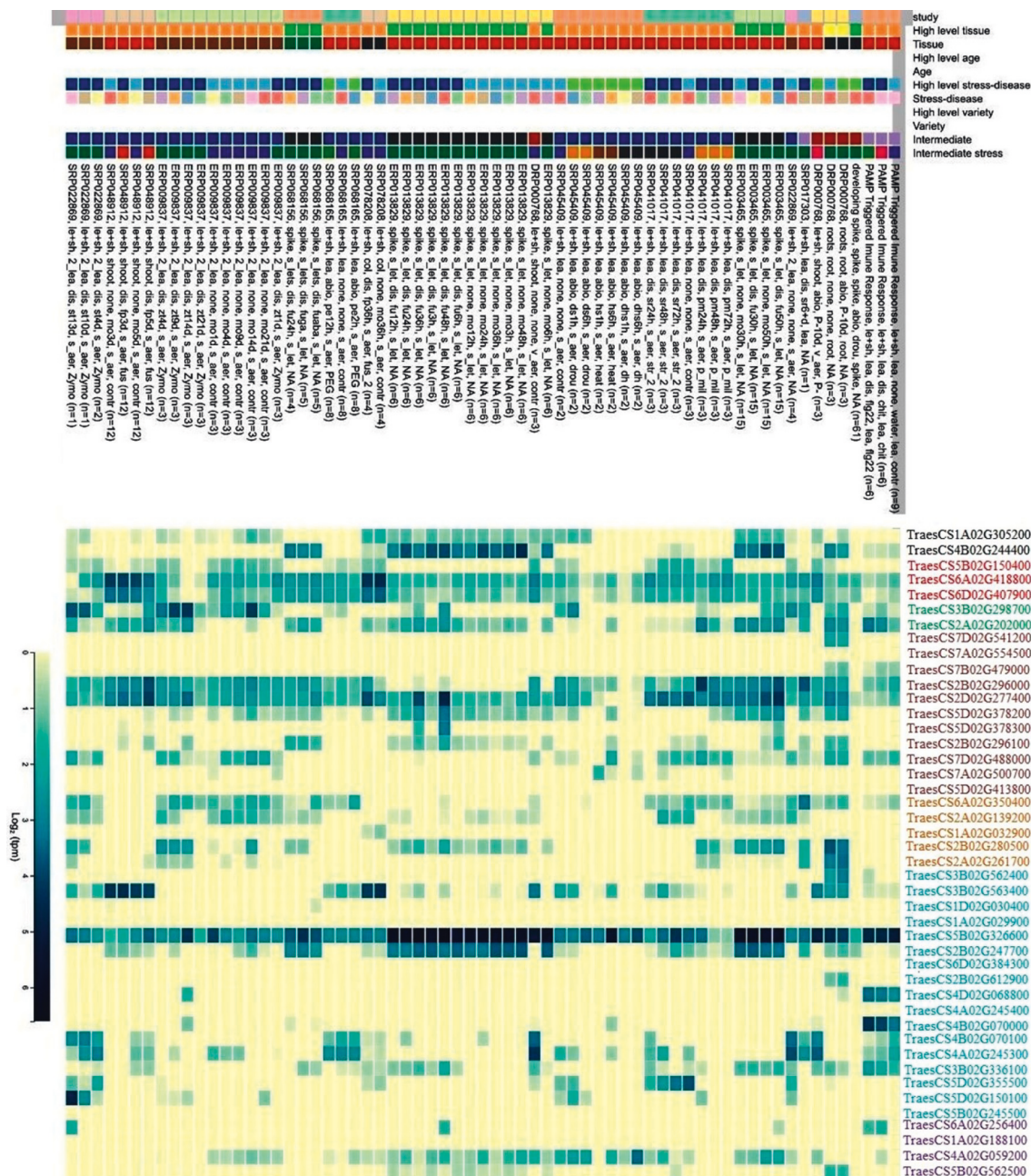


FIGURE 5: Heat map of MATE genes expressed in different degrees.

monogenic and oligogenic illnesses in medical genetics. It offers the potential to speed up medical sequencing investigations looking for genes causing Mendelian disorders or containing uncommon alleles causing complex features. It has been established that all physiological functions of life, including cell communication, are dependent on protein-protein interactions [31]. A simulation demonstrated that arsenic-induced MATE genes in rice were linked to 37 other genes [43]. Furthermore, roughly 30 genes were discovered in rice (*Oryza sativa* ssp. *japonica*) as part of the abiotic stress-sensitive gene sequence, which is engaged in stress-response signaling under diverse abiotic circumstances such as drought, submergence, cold, salt, and metal poisoning.

Twenty-two of these arsenic-induced MATE genes in rice were linked to 37 members, according to a computer simulation [43, 48]. Furthermore, around 30 genes were found in rice (*Oryza sativa* ssp. *japonica*) as part of the abiotic stress responder gene sequence, which also is engaged in strain communication under diverse abiotic drought conditions, saturation diving, cold, salt, and metal cytotoxicity. Using the STRING online tool, it was determined that 22 of the 30 seed meals and also the additional 34 generated neighbors were substantially engaged in the interacting proteins networks [49]. Using the STRING web tool, 30 seed proteins and 34 derived partners were discovered to be significantly engaged in the protein-protein

interaction structure [49]. Protein motifs have a specialized purpose in chemistry or biology [50, 51]. Despite the evolution, these short signature sequences persist in various proteins involved in common tasks [52]. The evolutionary tree's seven subfamilies were used to organize the wheat MATE protein's identified motifs. Protein motifs were shown to be comparable in groups that were genetically related. The motif compositions of tightly linked members of the same gene were similar [40, 53]. Studies of MATE transporters in soybean have been carried out by Liu [16], who used the MEME suite to discover that the first three families of MATE transporters have at least 12 conserved motif sequences, whereas the fourth family has considerably fewer motif sequences. Identifying different motifs in similarly functional genes enlightens the fundamentals of wheat functional genomics.

Wheat is an allopolyploid with two or three homologous subgenomes (two homoeoalleles in tetraploid wheat and hexaploid wheat has three homoeoalleles). Homoeoalleles of a genome in polyploid wheat are much more similar in DNA sequence as well as functionality, rendering gene cloning and recent experiments more challenging [54]. Understanding how homoeologous genes interact for cumulative expression was critical for identifying a strategy for crop improvement that included targeting and modifying one or more homoeologs [33, 39]. According to another study, toxin effluxes and vacuolar sequestration of GrMATE54, GrMATE53, and GaMATE21 were strongly expressed during abiotic stress conditions in cotton (Lu et al. 2018). In the leaves and roots of *Medicago truncatula*, the expression of UGT78G1, MaT4, MaT5, and MATE2 was identified [55, 56]. The most prevalent manifestations were found to be leaf, seed, and flower morphology; rosette organization pattern; and blooming period [57, 58]. Because of wheat's genomic complexity, a ternary plot was needed to analyze expression potentials under stress conditions. Researchers used ternary plot diagrams to estimate the impact of climate change on *Aspergillus flavus* and its aflatoxin B1 production in a similar investigation on *A. flavus* [26, 59]. Surprisingly, 70% of A, B, and D homoeologs determine reasonably with some other homoeologs and were generally released, whereas 30% demonstrated in an unbalanced fashion that was tissue-specific [60, 61].

5. Conclusion

Given the genetic concurrence research results, plant scientists are advised to swiftly design new cultivars more resistant to unforeseen stresses since MATE genes may be considered life-sustaining tools for the species they study. When genes from the same subfamily were subjected to biotic and abiotic stresses, their expression patterns varied dramatically. As a result, the promoter regions of these genes must be thoroughly investigated to see how they function independently. We have found that MATE genes were examined for their phylogenetic classification, protein-protein interactions, structural and functional analysis, protein motifs, and in silico expression analysis in the current study. With the use of Clustal Omega and phylogeny, a typical

phylogenetic tree was built utilizing the MATE genes, which had been further divided into seven families. The bulk of MATE genes were produced under biotic stress conditions produced by pathogen infections, with the genotype TraesCS5B02G326600.1 from family 1 having the highest expression profile after 4 days of inoculation under *Fusarium graminearum* infection of *Fusarium* head blight. To show variable expression patterns during biotic and abiotic stress scenarios, 39 ternary plots comprised genetically similar genetic makeup for 39 MATE genes were created using the wheat expression browsing tool. As a result, MATE transporters may be excellent candidates for training attempts to enhance agronomic features. The groundwork to speed up gene identification in polyploid wheat has finally been built after 5 years of wheat genomic resources being in flux. This foundation offers exciting chances to quicken wheat development and contribute to long-term food production security.

Data Availability

The data are available from the corresponding author upon request.

Conflicts of Interest

The authors declare that they have no conflicts of interest.

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References

- [1] Z. Han, Y. Liu, X. Deng et al., "Genome-wide identification and expression analysis of expansin gene family in common Wheat (*Triticum aestivum* L.)," *BMC Genomics*, vol. 20, no. 1, pp. 1–19, 2019.
- [2] B. S. Gill, R. Appels, A. M. Botha-Oberholster et al., "A workshop report on wheat genome sequencing: international genome research on wheat consortium," *Genetics*, vol. 168, no. 2, pp. 1087–1096, 2004.
- [3] R. Brechley, M. Spannagl, M. Pfeifer et al., "Analysis of the bread wheat genome using whole-genome shotgun sequencing," *Nature*, vol. 491, no. 7426, pp. 705–710, 2012.
- [4] G. R. Lazo, S. Chao, D. D. Hummel et al., "Development of an expressed sequence tag (EST) resource for Wheat (*Triticum aestivum* L.): EST generation, unigene analysis, probe selection, and bioinformatics for a 16,000-locus bin-delineated map," *Genetics*, vol. 168, no. 2, pp. 585–593, 2004.
- [5] K. Fan, M. Wang, Y. Miao et al., "Molecular evolution and expansion analysis of the NAC transcription factor in *Zea mays*," *PLoS One*, vol. 9, no. 11, Article ID e111837, 2014.

- [6] X. Shi and H. Q. Ling, "Current advances in genome sequencing of common wheat and its ancestral species," *The Crop Journal*, vol. 6, no. 1, pp. 15–21, 2018.
- [7] M. Kumar, M. Gouw, S. Michael et al., "ELM-the eukaryotic linear motif resource in 2020," *Nucleic Acids Research*, vol. 48, no. D1, pp. D296–D306, 2020.
- [8] T. Shoji, "ATP-binding cassette and multidrug and toxic compound extrusion transporters in plants: a common theme among diverse detoxification mechanisms," *Int Rev Cell Mol Biol*, vol. 309, pp. 303–346, 2014.
- [9] X. Liu, Z. Liu, X. Niu, Q. Xu, and L. Yang, "Genome-wide identification and analysis of the NPR1-like gene family in bread wheat and its relatives," *International Journal of Molecular Sciences*, vol. 20, no. 23, p. 5974, 2019.
- [10] K. Takanashi, N. Shitan, and K. Yazaki, "The multidrug and toxic compound extrusion (MATE) family in plants," *Plant Biotechnology J.* vol. 14, Article ID 0904, 2014.
- [11] M. H. Brown, I. T. Paulsen, and R. A. Skurray, "The multidrug efflux protein NorM is a prototype of a new family of transporters," *Molecular Microbiology*, vol. 31, no. 1, pp. 394–395, 1999.
- [12] Y. Morita, K. Kodama, S. Shiota et al., "NorM, a putative multidrug efflux protein, of *Vibrio parahaemolyticus* and its homolog in *Escherichia coli*," *Antimicrobial Agents and Chemotherapy*, vol. 42, no. 7, pp. 1778–1782, 1998.
- [13] A. C. Diener, R. A. Gaxiola, and G. R. Fink, "Arabidopsis ALF5, a multidrug efflux transporter gene family member, confers resistance to toxins," *The Plant Cell Online*, vol. 13, no. 7, pp. 1625–1638, 2001.
- [14] L. Wang, X. Bei, J. Gao, Y. Li, Y. Yan, and Y. Hu, "The similar and different evolutionary trends of the MATE family occurred between rice and *Arabidopsis thaliana*," *BMC plant boil*, vol. 16, no. 1, pp. 1–19, 2016.
- [15] A. L. Dos Santos, S. Chaves-Silva, L. Yang et al., "Global analysis of the MATE gene family of metabolite transporters in tomato," *BMC Plant Biology*, vol. 17, no. 1, pp. 1–13, 2017.
- [16] J. Liu, Y. Li, W. Wang, J. Gai, and Y. Li, "Genome-wide analysis of MATE transporters and expression patterns of a subgroup of MATE genes in response to aluminum toxicity in soybean," *BMC Genomics*, vol. 17, no. 1, pp. 1–15, 2016.
- [17] P. Lu, R. O. Magwanga, X. Guo et al., "Genome-wide analysis of multidrug and toxic compound extrusion (MATE) family in *Gossypium raimondii* and *Gossypium arboreum* and its expression analysis under salt, cadmium, and drought stress," *G3: Genes, Genomes, Genetics*, vol. 8, no. 7, pp. 2483–2500, 2018.
- [18] G. Chen, H. Liang, Q. Zhao, A. M. Wu, and B. Wang, "Exploiting MATE efflux proteins to improve flavonoid accumulation in *Camellia sinensis* in silico," *International Journal of Biological Macromolecules*, vol. 143, pp. 732–743, 2020.
- [19] L. Chen, Y. Liu, H. Liu et al., "Identification and expression analysis of MATE genes involved in flavonoid transport in blueberry plants," *PLoS One*, vol. 10, no. 3, Article ID e0118578, 2015.
- [20] R. Appels, K. Eversole, N. Stein et al., "Shifting the limits in wheat research and breeding using a fully annotated reference genome," *Science*, vol. 17, p. 361(6403), 2018.
- [21] G. D. Stormo, "An introduction to sequence similarity ("homology") searching," *Current Protocols in Bioinformatics*, vol. 27, no. 1, pp. 3–1, 2009.
- [22] S. Ajay Kumar, I. Raza Khan, S. Khan, K. Pant, S. Debnath, and S. Miah, "Multichannel CNN model for biomedical entity reorganization," *BioMed Research International*, 2022.
- [23] R. Apweiler, A. Bairoch, C. H. Wu et al., "UniProt: the universal protein knowledgebase," *Nucleic Acids Research*, vol. 32, no. suppl 1, pp. D115–D119, 2004.
- [24] M. K. Tello-Ruiz, S. Naithani, J. C. Stein et al., "Gramene 2018: unifying comparative genomics and pathway resources for plant research," *Nucleic Acids Research*, vol. 46, no. D1, pp. D1181–D1189, 2018.
- [25] D. T. Hoang, O. Chernomor, A. von Haeseler, B. Q. Minh, and L. S. Vinh, "UFBoot2: improving the ultrafast bootstrap approximation," *Molecular Biology and Evolution*, vol. 35, no. 2, pp. 518–522, 2018.
- [26] J. Trifinopoulos, L. T. Nguyen, A. von Haeseler, and B. Q. Minh, "W-IQ-TREE: a fast online phylogenetic tool for maximum likelihood analysis," *Nucleic Acids Research*, vol. 44, no. W1, pp. W232–W235, 2016.
- [27] H. Li, M. Shabaz, and R. Castillejo-Melgarejo, "Implementation of python data in online translation crawler website design," in *International Journal of System Assurance Engineering and Management* Springer Science and Business Media LLC, 2021.
- [28] J. S. Papadopoulos and R. Agarwala, "COBALT: a constraint-based alignment tool for multiple protein sequences," *Bioinformatics*, vol. 23, no. 9, pp. 1073–1079, 2007.
- [29] X. Peng, Y. Zhao, J. Cao et al., "CCCH-type zinc finger family in maize: genome-wide identification, classification and expression profiling under abscisic acid and drought treatments," *PLoS One*, vol. 7, no. 7, Article ID e40120, 2012.
- [30] A. Singh, I. Khan, S. Khan, K. Pant, S. Debnath, and S. Miah, "Multichannel CNN Model for biomedical entity reorganization," *BioMed Research International*, vol. 2022, Article ID 5765629, 2022.
- [31] D. Szklarczyk, J. H. Morris, H. Cook et al., "The STRING database in 2017: quality-controlled protein-protein association networks, made broadly accessible," *Nucleic Acids Research*, vol. gkw937, 2016.
- [32] M. Sivaguru, J. Liu, and L. V. Kochian, "Targeted expression of Sb MATE in the root distal transition zone is responsible for sorghum aluminum resistance," *The Plant Journal*, vol. 76, no. 2, pp. 297–307, 2013.
- [33] P. Borrill, N. Adamski, and C. Uauy, "Genomics as the key to unlocking the polyploid potential of Wheat," *New Phytologist*, vol. 208, no. 4, pp. 1008–1022, 2015.
- [34] P. Borrill, R. Ramirez-Gonzalez, and C. Uauy, "expVIP: customizable RNA-seq data analysis and visualization platform," *Plant Physiology*, vol. 170, no. 4, pp. 2172–2186, 2016.
- [35] W. R. Pearson, *An Introduction to Sequence Similarity ("homology") Searching*, CurrProtocBioinformatics, 2013.
- [36] L. L. Qi, B. Echallier, S. Chao et al., "A chromosome bin map of 16,000 expressed sequence tag loci and distribution of genes among the three genomes of polyploid wheat," *Genetics*, vol. 168, pp. 701–712, 2004.
- [37] H. Ma, H. Zhao, Z. Liu, and J. Zhao, "The phytoeyanin gene family in rice (*Oryza sativa* L.): genome-wide identification, classification, and transcriptional analysis," *PLoS One*, vol. 6, no. 10, Article ID e25184, 2011.
- [38] T. L. Bailey, J. Johnson, C. E. Grant, and W. S. Noble, "The MEME suite," *Nucleic Acids Research*, vol. 1:43, no. W1, pp. W39–W49, 2015.
- [39] T. L. Bailey and C. Elkan, "Unsupervised learning of multiple motifs in biopolymers using expectation maximization," *Machine Learning*, vol. 21, no. 1, pp. 51–80, 1995.
- [40] H. Zhu, J. Wu, Y. Jiang et al., "Genomewide analysis of MATE-type gene family in maize reveals microsynteny and

- their expression patterns under aluminum treatment," *Journal of Genetics*, vol. 95, no. 3, pp. 691–704, 2016.
- [41] T. K. Lohani, M. T. Ayana, A. K. Mohammed, M. Shabaz, G. Dhiman, and V. Jagota, "A comprehensive approach of hydrological issues related to ground water using GIS in the Hindu holy city of Gaya, India," *World Journal of Engineering*, 2021.
 - [42] Y. M. Bi, S. Kant, J. Clark et al., "Increased nitrogen-use efficiency in transgenic rice plants over-expressing a nitrogen-responsive early nodulin gene identified from rice expression profiling," *Plant, Cell and Environment*, vol. 32, no. 12, pp. 1749–1760, 2009.
 - [43] S. Seth, S. Debnath, and N. R. Chakraborty, "In silico analysis of functional linkage among arsenic-induced MATE genes in rice," *Biotechnol. Rep.* vol. 26, Article ID e00390, 2020.
 - [44] R. H. Ramírez-González, P. Borrill, D. Lang et al., "The transcriptional landscape of polyploid wheat," *Science*, vol. 361, no. 6403, 2018.
 - [45] K. P. Rao, T. A. Richa, K. U. Kumar, B. A. Raghuram, and A. K. Sinha, "In silico analysis reveals 75 members of mitogen-activated protein kinase kinase gene family in rice," *DNA Research*, vol. 17, no. 3, pp. 139–153, 2010.
 - [46] M. Gaur, A. Tiwari, R. P. Chauhan, D. Pandey, and A. Kumar, "Molecular modeling, docking and protein-protein interaction analysis of MAPK signaling cascade involved in Camalexin biosynthesis in *Brassica rapa*," *Bioinformation*, vol. 14, no. 4, p. 145, 2018.
 - [47] S. Soyk, M. Benoit, and Z. B. Lippman, "New horizons for dissecting epistasis in crop quantitative trait variation," *Annual Review of Genetics*, vol. 54, pp. 287–307, 2020.
 - [48] S. Gupta, J. A. Stamatoyannopoulos, T. L. Bailey, and W. S. Noble, "Quantifying similarity between motifs," *Genome Biology*, vol. 8, no. 2, pp. 1–9, 2007.
 - [49] P. Muthuramalingam, S. R. Krishnan, R. Pothiraj, and M. Ramesh, "Global transcriptome analysis of combined abiotic stress signaling genes unravels key players in *Oryza sativa* L.: an in silico approach," *Frontiers of Plant Science*, vol. 8, p. 759, 2017.
 - [50] D. Conklin, "Machine discovery of protein motifs," *Machine Learning*, vol. 21, no. 1, pp. 125–150, 1995.
 - [51] R. Khan, M. Shabaz, S. Hussain, F. Ahmad, and P. Mishra, "Early flood detection and rescue using bioinformatic devices, internet of things (IOT) and Android application," *World Journal of Engineering*, 2021.
 - [52] C. G. Nevill-Manning, T. D. Wu, and D. L. Brutlag, "Highly specific protein sequence motifs for genome analysis," *Proceedings of the National Academy of Sciences*, vol. 95, no. 11, pp. 5865–5871, 1998.
 - [53] C. Zhou, Z. Dong, T. Zhang et al., "Genome-scale analysis of homologous genes among subgenomes of bread wheat (*Triticum aestivum* L.)," *International Journal of Molecular Sciences*, vol. 21, no. 8, p. 3015, 2020.
 - [54] Y. Zhang, D. Li, D. Zhang et al., "Analysis of the functions of Ta GW 2 homoeologs in wheat grain weight and protein content traits," *The Plant Journal*, vol. 94, no. 5, pp. 857–866, 2018.
 - [55] T. L. Bailey and C. Elkan, "Fitting a mixture model by expectation maximization to discover motifs in biopolymers," in *Proceedings of the Second International Conference on Intelligent Systems for Molecular Biology*, pp. 28–36, AAAI Press, Menlo Park, California, 1994.
 - [56] J. Zhao, D. Huhman, G. Shadle et al., "MATE2 mediates vacuolar sequestration of flavonoid glycosides and glycoside malonates in *Medicago truncatula*," *The Plant Cell Online*, vol. 23, no. 4, pp. 1536–1555, 2011.
 - [57] T. L. Bailey, "DREME: motif discovery in transcription factor ChIP-seq data," *Bioinformatics*, vol. 27, no. 12, pp. 1653–1659, 2011.
 - [58] M. Tiwari, D. Sharma, M. Singh, R. D. Tripathi, and P. K. Trivedi, "Expression of OsMATE1 and OsMATE2 alters development, stress responses and pathogen susceptibility in *Arabidopsis*," *Scientific Reports*, vol. 4, no. 1, pp. 1–12, 2014.
 - [59] A. Medina, A. Rodriguez, and N. Magan, "Effect of climate change on *Aspergillus flavus* and aflatoxin B1 production," *Frontiers in Microbiology*, vol. 5, p. 348, 2014.
 - [60] J. F. Uhrig, "Protein interaction networks in plants," *Planta*, vol. 224, no. 4, pp. 771–781, 2006.
 - [61] UniProt Consortium, "UniProt: the universal protein knowledgebase in 2021," *Nucleic Acids Research*, vol. 49, no. D1, pp. D480–D489, 2021.